

Problem statement

Measuring Universal Meaningful Connectivity in a hyper-connected city

Seoul — a hard case for testing UMC.

Seoul ranks among the world's most connected cities by every conventional ITU benchmark — yet conventional **threshold-based UMC frameworks** cannot distinguish meaningful variation in a city where basic infrastructure is already saturated.

Any remaining gaps are **structural and human, not infrastructural**. Seoul is therefore the natural setting for asking how UMC should be defined and measured beyond infrastructure.

9.4M

Population

25

Autonomous districts

99%+

Broadband penetration

100%

5G coverage

Q1 — MEASUREMENT

How should UMC be measured in **high-connectivity urban contexts** where threshold-based frameworks may not capture meaningful variation?

Q2 — MECHANISM

Are certain groups **structurally excluded** from connectivity benefits, and what methodology can separate individual vulnerability from district context?

SIX UMC DIMENSIONS

- Connectivity
- Available for Use
- Affordability
- Devices
- Digital Skills
- Safety

Data sources

Analysis 1 & 2

Dataset name	Source	Link	Coverage	Key variables	Format	Access method
Seoul Digital Competency Survey	Seoul AI Foundation	https://saif.or.kr/dataroom/2564	Geographic: Seoul Time: 2023 used	devices, skills, safety; survey weights	CSV/XLSX	data provision upon official request
Cellular Base Station Data	Spectrum Resource Mgmt System	https://spectrummap.kr/gis/mobile_service.do?menuNo=300480	Geographic: Seoul Time: 2023, 2024	4G/5G cell-site locations		Public Open Data
NIA Communication Quality (CQ)	Korean Public Data Portal	https://www.data.go.kr/data/15120591/fileData.do	Geographic: Seoul Time: 2023, 2024	district-level download speed		
Public Free Wi-Fi Installation Data	Local Admin WiFi License Data	https://www.localdata.go.kr/lif/lifeCtacDataView.do	Geographic: Seoul Time: 2023, 2024	public Wi-Fi installation locations		
Telecommunication Indicators	SKT Telecom Data	https://www.data.go.kr/data/15127168/fileData.do#	Geographic: Seoul Time: 2023, 2024	mobile data use; online service days; fee delinquency rate		
Average Monthly Household Income by District	Seoul Commercial Analysis Service	https://data.seoul.go.kr/dataList/OA-22167/S/1/datasetView.do	Geographic: Seoul Time: 2023, 2024	average monthly household income		
Seoul Survey	Seoul Open Data	https://data.seoul.go.kr/dataList/OA-15564/F/1/datasetView.do	Geographic: Seoul Time: 2023, 2024	digital service use; demographics, income, housing, district/year		

Analysis 3

Dataset name	Source	Link	Coverage	Key variables	Format	Access method
Daangn Community Post	Daangn Market	daangn.com/kr/community/s	Geographic: Seoul dong fields Time: 2024-2026 crawl period	post text; gu/location; createdAt; subject/category	CSV/XLSX	Web Scraping
Seoul Living/Floating Population	Seoul Open Data	data.seoul.go.kr OA-14991	Geographic: Seoul Time: 2023-2026 used as available	hourly/dong population; daytime population; age shares; dong/district code		Public Open Data

Stage 1 — UMC index construction

6 dimensions × 25 districts, equal-weighted composite



KEY METHODOLOGICAL CHOICES

Daytime population as denominator

Connectivity indicators use daytime (floating) population rather than registered residents — captures where people actually access digital services.

Reverse-coded burden indicators

Telecom bill arrears rate is reverse-coded so higher values consistently mean better outcomes across all six dimensions.

Equal weights across dimensions

Composite UMC score uses equal weights across the six dimensions — no a priori prioritization, supporting reproducible cross-city application.

Stage 2 (HLM) & Stage 3 (Platform text)

Multilevel modeling and Bayesian text-evidence integration

STAGE 2 — HIERARCHICAL LINEAR MODEL

Dependent variable

Digital-service usage score (0–80) summed from 8 items: finance, e-commerce, public services, communication, leisure, education, mobile payment.

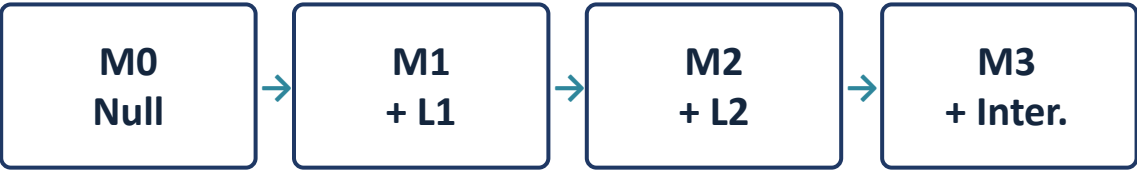
Level-1 predictors (individual)

Age · gender · education · income · occupation · housing type · disability · single-person household (following prior digital divide literature; van Dijk, 2020; Mossberger et al., 2006)

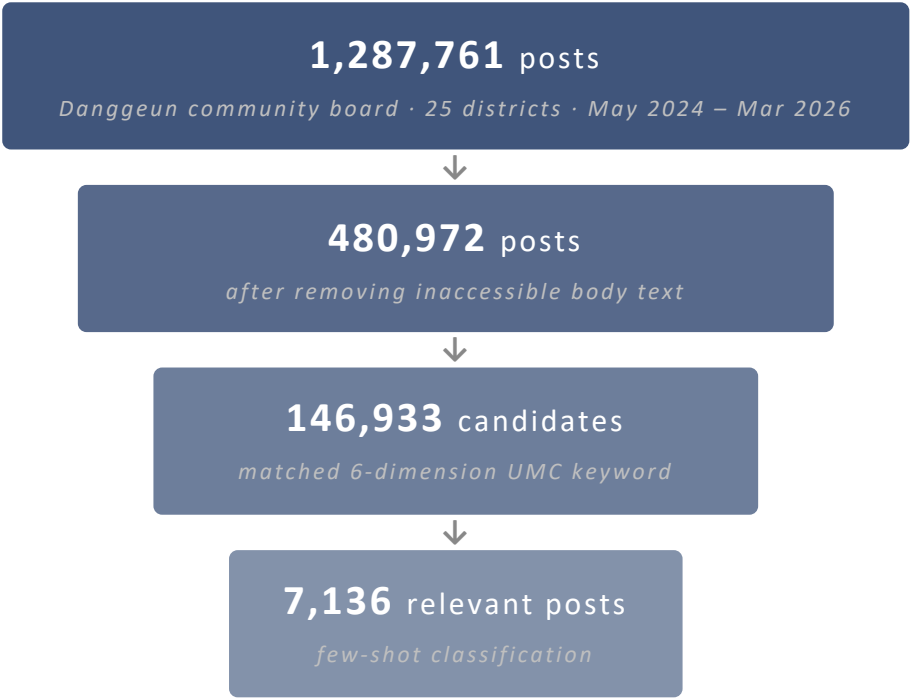
Level-2 predictors (district)

Connectivity · AFU · Affordability (from Stage 1).

Sequential model estimation

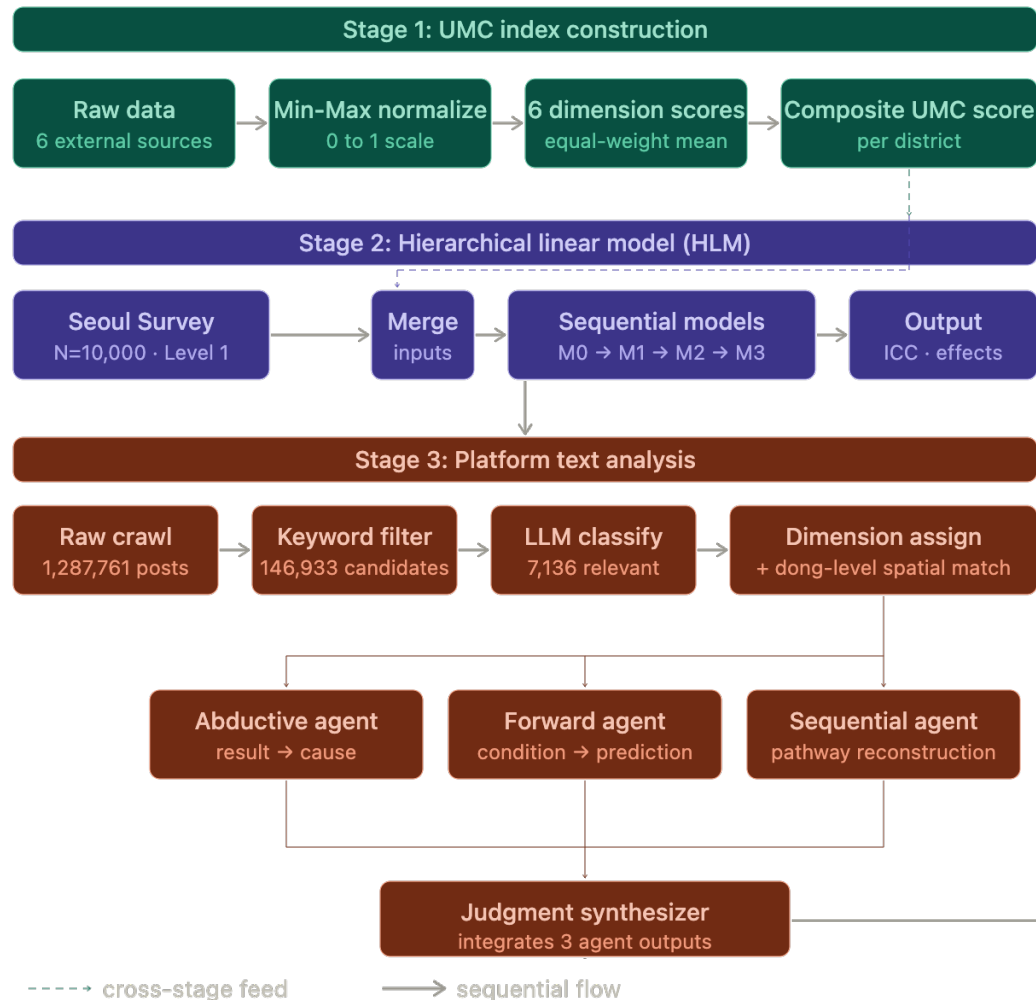


STAGE 3 — PLATFORM TEXT ANALYSIS



Integrated ETL pipeline

Three sequential stages with cross-stage feedback loop



STAGE INTEGRATION LOGIC

Stage 1 establishes *where* digital gaps exist

→ district-level structural baseline

Stage 2 explains *who* is affected and *why*

→ individual vulnerabilities tested against that baseline

Stage 3 reveals *how* residents experience those gaps

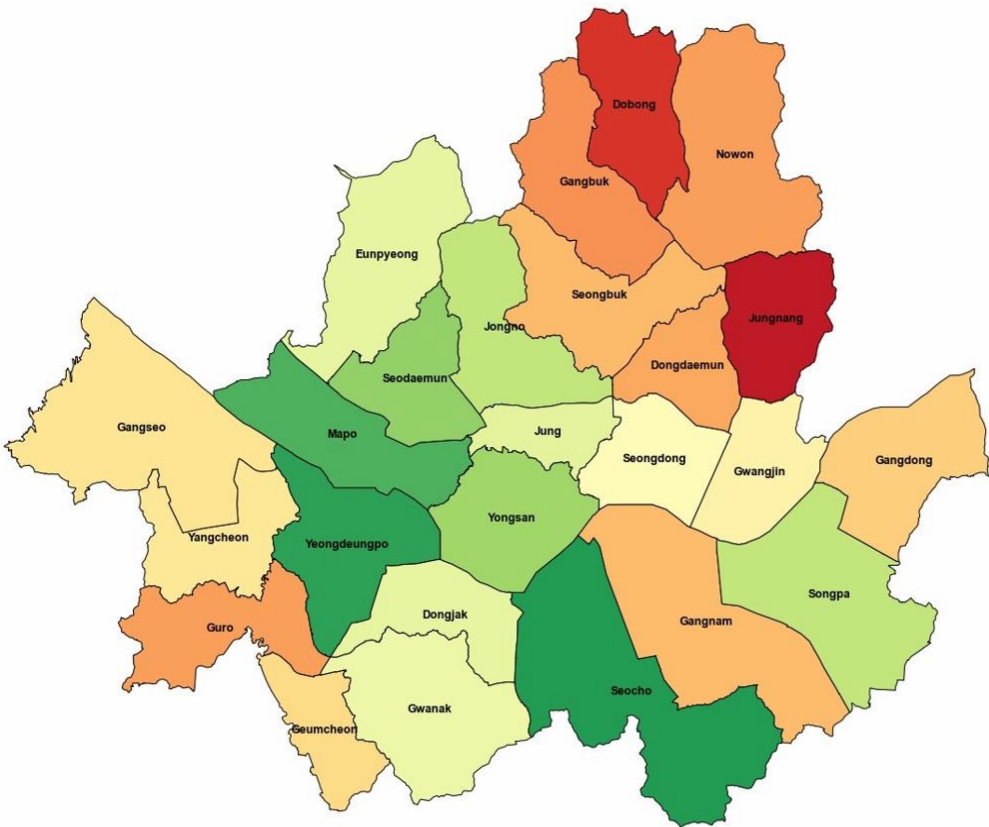
→ platform signals validate and nuance the quantitative picture

Closed loop: Stage 1 scores re-enter Stage 3 as Bayesian priors

— measurement informs interpretation

Preliminary results — Where is the gap?

UMC Composite Index across 25 districts



UMC Composite Score by district, Seoul 2024.

TOP 5 DISTRICTS

1. Seocho-gu	0.695
2. Yeongdeungpo-gu	0.686
3. Mapo-gu	0.665
4. Seodaemun-gu	0.615
5. Yongsan-gu	0.603

BOTTOM 5 DISTRICTS

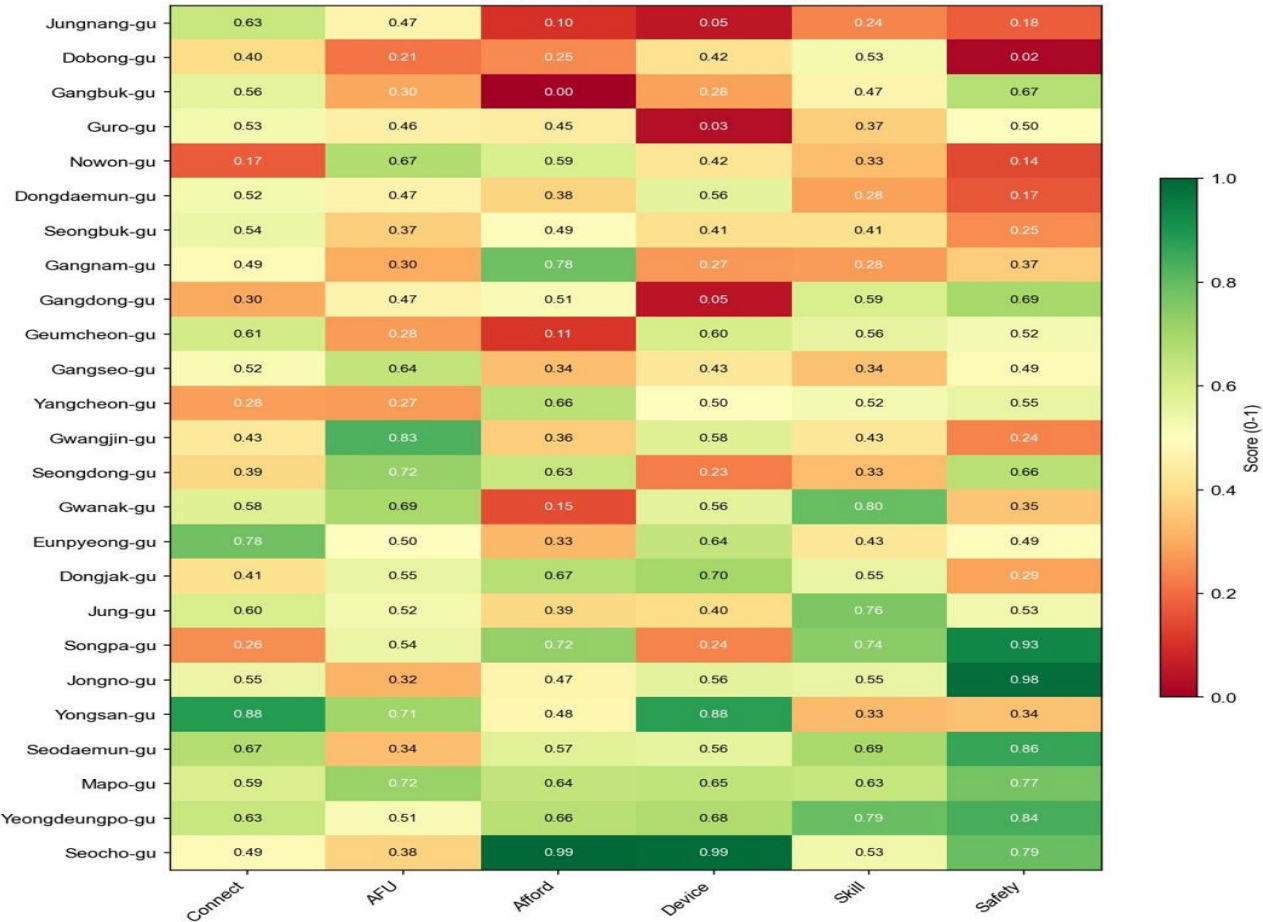
21. Nowon-gu	0.389
22. Guro-gu	0.389
23. Gangbuk-gu	0.380
24. Dobong-gu	0.303
25. Jungnang-gu	0.279

KEY FINDINGS

- 2.5× gap between top and bottom districts
- Northern Seoul (Jungnang, Dobong, Gangbuk, Nowon) forms a Low-Low spatial cluster
- Spatial autocorrelation confirms statistically significant clustering

Dimensional heterogeneity

No district excels in all dimensions



Heatmap of district x UMC dimension scores, Seoul 2024.

ILLUSTRATIVE CASES

Eunpyeong-gu

- Connectivity **0.78** *top tier*
- Affordability **0.33** *bottom*

Gwanak-gu

- Digital Skills **0.80** *#1 in Seoul*
- Affordability **0.15** *last*

KEY FINDINGS

- Digital deserts have average connectivity but severe gaps in Devices and Safety
- Physical infrastructure exists — what is missing is human and economic capacity
- A single composite ranking hides policy-relevant, dimension-specific vulnerabilities
- Implication: dimension-specific intervention, not uniform infrastructure investment

Why does the gap exist? — Individual differences dominate

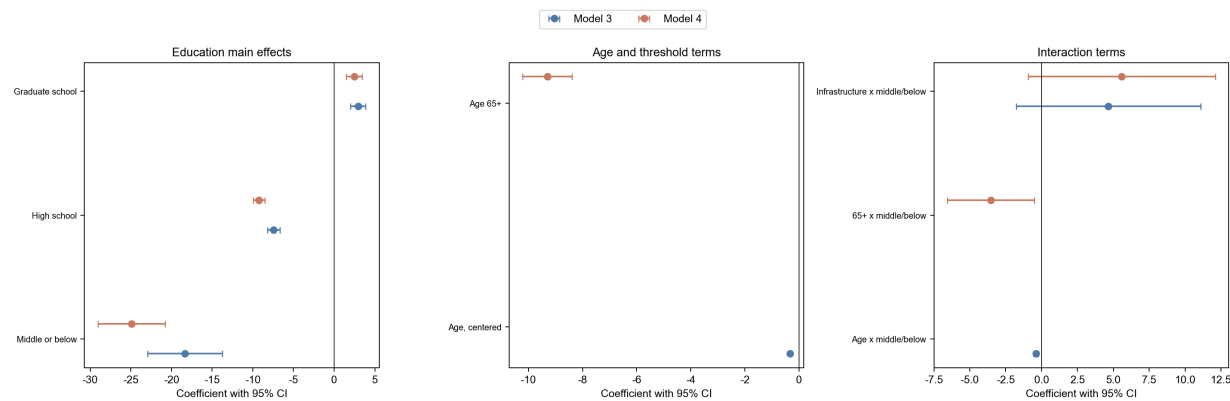
HLM: ICC and main effects

WHERE THE GAP LIVES

Almost all variation in digital use sits between **people**, not between **districts**.

STRONGEST PREDICTORS

Edu ≤ middle school	-18.4 pts	$p<0.001$
Age	-0.34 / yr	$p<0.001$
Single-person household	-1.7 pts	$p<0.001$
Disabled	+3.1 pts	$p<0.001$
Detached housing	-2.4 pts	$p<0.001$
Infrastructure main effect	≈ 0	$n.s.$



Model 3 vs Model 4

Does infrastructure offset individual vulnerability?

HLM: cross-level interactions

MAIN EFFECT

District-level UMC dimensions show only weak, contextual associations with individual digital use.

CROSS-LEVEL INTERACTIONS

Connectivity × low education	+4.66 n.s. (M3)
Age × low education	−0.40 *** (M3)
Older (65+) × low education	−3.53 * (M4 supp.)
L2 Affordability (main)	−1.35 † (M3)

INTERPRETATION

Retained interactions are individual-level: the low-education penalty steepens with age (Age × low education −0.40***, M3) and with being 65+ (Older × low education −3.53*, M4 supplementary). District-level UMC dimensions show only weak, marginal main associations (Connectivity −1.34†, Affordability −1.35† in M3); the Connectivity × low-education term is positive but non-significant (+4.66, p=.16). District scores locate where vulnerable residents concentrate — supporting place-sensitive (not place-based) targeting.

Cross-level interactions: how district-level UMC dimensions interact with vulnerable groups.

Where infrastructure indicators may miss the problem

Bayesian posterior shifts: Stage 1 priors updated by platform text evidence

METHOD

UMC dimension scores serve as Bayesian priors; classified Danggeun posts update them. The resulting **posterior shifts** reveal where measured indicators may overstate or understate lived deprivation.

INTERPRETATION

POSITIVE shift → measurement misses deprivation that residents actually express on the platform

NEGATIVE shift → measurement overstates deprivation relative to lived signals

Across 150 cells, the dominant pattern (preliminary): Skills and Safety show hidden deprivation in northern districts; AFU and Affordability tend to be overstated by measurement.

1.29M

Posts crawled
(2024–2026)

~520k

Analyzed after
dedup & filtering

25

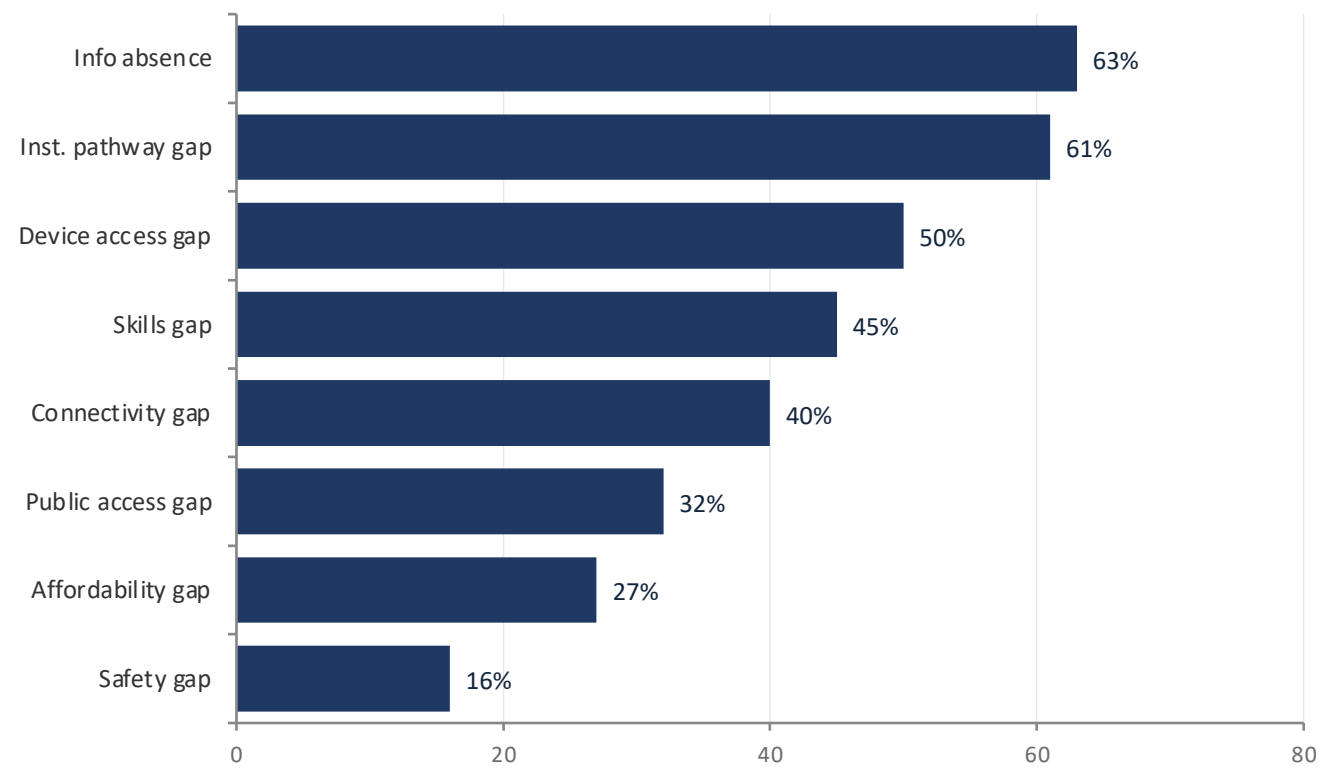
Districts covered
(all of Seoul)

PRELIMINARY POSTERIOR SHIFTS

Jungnang · Devices	+4.77 pp
<i>measured score understates device-access deprivation</i>	
Jungnang · Skills	+4.69 pp
<i>platform signals stronger skill-related struggles</i>	
Gwanak · Skills	−4.52 pp
<i>measured score overstates skill deprivation</i>	
Yongsan · Devices	−4.18 pp
<i>platform signals fewer device-related issues</i>	

How is the gap experienced?

Platform text analysis — deprivation patterns from ~520k community posts.



Deprivation Types (n=100) — 88% of posts overlap 2 or more types.

REPRESENTATIVE CASES

Sinsa-dong, Eunpyeong-gu

“Is there really no place...?”

INFO ABSENCE • DIGITAL SKILLS

Bongcheon-dong, Gwanak-gu

“Where can I print legal documents? Regular print shops can't do it.”

PUBLIC ACCESS GAP • AVAILABILITY FOR USE

Sillim-dong, Gwanak-gu

“Looking for a PC-bang that can do document scanning.”

DEVICE ACCESS GAP • DEVICES + AVAILABILITY

Progress status & next steps

Three analyses underway; Phase 2 of platform text validation in progress

✓ ANALYSIS COMPLETE Analysis 1 · UMC Index • 25 districts × 6 dimensions (2023, 2024) • Composite + dimensional scores finalized • Spatial autocorrelation validated Validated against external benchmarks.	🕒 MAIN RESULTS DRAFTED Analysis 2 · HLM • Models M0–M3 estimated • ICC, fixed effects, cross-level interactions complete • Sensitivity checks pending finalization Preliminary — final sensitivity & dong-level robustness in progress.	🕒 PHASE 2 IN PROGRESS Analysis 3 · Multi agent + Bayesian • Phase 0 – 1 complete: classification, hypothesis generation (n=100) • Initial posterior shifts computed on validation sample • Phase 2 (judge-agent triangulation on full corpus) running Preliminary — values may shift after Phase 2 validation.
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NEXT STEPS BEFORE FINAL DELIVERABLE

- Phase 2 judge-agent triangulation on the full corpus of LLM-relevant posts
- Final policy implications: refine the four preliminary directions with validated evidence